

torch.cuda.comm.gather#

<https://pytorch.org/docs/stable/generated/torch.cuda.comm.gather.html#torch.cuda.comm.gather>

Description:

Gathers tensors from multiple GPU devices. tensors (Iterable[Tensor]) – an iterable of tensors to gather. Tensor sizes in all dimensions other than dim have to match. dim (int, optional) – a dimension along which the tensors will be concatenated. Default: 0. destination (torch.device, str, or int, optional) – the output device. Can be CPU or CUDA. Default: the current CUDA device. out (Tensor, optional, keyword-only) – the tensor to store gather result. Its sizes must match those of tensors, except for dim, where the size must equal $\sum(\text{tensor.size}(\text{dim}) \text{ for tensor in tensors})$. Can be on CPU or CUDA.

Function Signature

```
torch.cuda.comm.gather(tensors, dim=0, destination=None, *,  
out=None)[source]#
```

Parameters

Returns

If destination is specified,

If out is specified,

Returns:

If destination is specified, a tensor located on destination device, that is a result of concatenating tensors along dim. If out is specified, the out tensor, now containing results of concatenating tensors along dim.

Notes & Warnings

NOTE

Note destination must not be specified when out is specified.

Detailed Documentation

Documentation

Gathers tensors from multiple GPU devices.

tensors (Iterable[`Tensor`]) – an iterable of tensors to gather. Tensor sizes in all dimensions other than dim have to match.

dim (int, optional) – a dimension along which the tensors will be concatenated. Default: 0.

destination (torch.device, str, or int, optional) – the output device. Can be CPU or CUDA. Default: the current CUDA device.

out (`Tensor`, optional, keyword-only) – the tensor to store gather result. Its sizes must match those of tensors, except for dim, where the size must equal $\sum(\text{tensor.size}(\text{dim}) \text{ for tensor in tensors})$. Can be on CPU or CUDA.

destination must not be specified when out is specified.

a tensor located on destination device, that is a result of concatenating tensors along dim.

the out tensor, now containing results of concatenating tensors along dim.